Buying Tokens, Not Feelings: Output-Format Constraints Reveal the Reasoning-Budget Driver of Affective Prompting

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Abstract

Affective and self-evaluative prompts are reported to improve large language model (LLM) accuracy, but the mechanism is unclear. We present a controlled ablation that pins it down. On an exact-scored cognitive-control battery, telling Qwen2.5-7B to feel uncertain and check its work raises accuracy by +29.2 percentage points, and, surprisingly, telling it that its answers are "often wrong" raises accuracy by +15.5 points. Yet the *same* prompts, on the *same* tasks and model, collapse to 15–19 points *below* baseline the moment their output is constrained to a final answer only. This is the central finding: prompt-induced gains are not robust to output-format constraints, with direct practical consequences wherever concise or structured outputs are required. We explain the pattern with a *reasoning-budget* account: a self-evaluative prompt helps only when it induces task-relevant visible computation, and only when the model has budget headroom to spend. Three lines of evidence support this. (1) Direct non-affective scaffolds match or beat self-doubt ("check your work" +37.4, "think step by step" +33.6, forced steps +41.8). (2) Concise variants of *every* effective prompt fall below baseline. (3) The gain vanishes for models that already reason at length: two 14B reasoning models that emit $\sim$450 default tokens show no self-doubt benefit, whereas the 7B model that emits $\sim$80 does. Across 32,400 paired mechanism rows we find the effect is sharply model-dependent, which we recast not as a limitation but as a map of when metacognitive prompts can help at all.

1 Introduction

Emotional and self-evaluative prompting is reported to improve large language model (LLM) performance. EmotionPrompt reports gains from psychologically framed task suffixes, and NegativePrompt reports similar gains from negative emotional stimuli [Li et al., 2023, Wang et al., 2024]. Recent evidence complicates the picture: static emotional prefixes often produce small, task-dependent effects rather than reliable general-purpose improvements [Zhao et al., 2026]. A central methodological question therefore remains open: when an affective or self-evaluative prompt improves an LLM, what has actually changed?

This article answers that question with a clean ablation rather than a new prompt. Our headline observation is a sharp *fragility*: prompts that produce large gains can be made to produce large *losses* by an output-format constraint alone, holding task and model fixed. On an exact-scored cognitive-control battery, instructing Qwen2.5-7B to feel uncertain and check its work raises accuracy by +29.2 percentage points (pp). Constraining that same model to give only a final answer drives it, on the same tasks, to 15.5 pp *below* baseline (the doubt+concise condition); a cleaner control that keeps the checking instruction verbatim but suppresses visible reasoning behaves the same way (Section 4.3). Whatever the prompt is doing, it is not changing what the model knows; it is changing

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how much task-relevant computation the model emits before answering. This matters in practice because concise and structured outputs are the norm in deployment (chat assistants with latency budgets, JSON/function-calling APIs, and extraction pipelines), precisely the settings in which these celebrated prompt gains silently disappear or invert.

We package this into a *reasoning-budget* account: a self-evaluative prompt improves accuracy only insofar as it induces additional task-relevant visible computation (checking, updating, explicit state tracking), and only when the model has headroom above its default budget to spend. The account makes three predictions, each of which we confirm. First, if the active ingredient is induced computation rather than affect, then direct non-affective scaffolds should match or beat affective ones; they do ("check your work" +37.4 pp, "think step by step" +33.6 pp, forced steps +41.8 pp, all $\geq$ self-doubt). Second, suppressing the visible computation should remove the gain regardless of the prompt's wording; concise variants of self-doubt, checking, and step-by-step all fall below baseline. Third, models that already spend a large default reasoning budget should show no gain because they have no headroom; two 14B reasoning models that emit $\sim$450 default tokens are flat or slightly negative under self-doubt, while the 7B model that emits $\sim$80 default tokens gains sharply.

The account also explains a result that is otherwise bizarre: telling Qwen2.5 that "your answers are often wrong and your reasoning is unreliable" *improves* its accuracy by +15.5 pp. Under a literal reading this should hurt. Under the reasoning-budget account it is expected: negative self-framing triggers compensatory checking, in which the model spends more tokens verifying before committing, rather than degrading competence. The same lens explains why the effect is strongly model-dependent. We therefore treat cross-model divergence not as a weakness but as a contribution: it is an empirical map of *when* affective and metacognitive prompts can help, governed by post-training-shaped instruction following and by how much budget headroom a model has.

We make four contributions. (1) We demonstrate the **output-format fragility** of prompt-induced gains with a paired ablation: holding the reasoning instruction fixed and removing only its visible expression flips a large gain into a large loss. (2) We give a **reasoning-budget account** that unifies the affective, metacognitive, capability-denial, and concise results, and a budget-control experiment that isolates visible computation as the mediating variable. (3) We show a **budget-ceiling boundary**: default-reasoning models show no benefit, giving a directional answer to the obvious "does it scale?" question. (4) We contribute a small, exact-scored cognitive-control battery and roughly 37,000 paired rows of open-weight model behavior (32,400 mechanism, 3,600 budget-control, and 1,200 pilot) under affective, metacognitive, and format-control prompts.

2 Related Work

Affective prompt engineering has treated emotional language as a task-level performance intervention. EmotionPrompt appends emotional stimuli to user prompts and reports benchmark improvements [Li et al., 2023]. NegativePrompt extends this idea using negative emotional stimuli and reports that negative framing can also improve performance [Wang et al., 2024]. A newer large-scale study finds that fixed emotional framing is usually mild, input-dependent, and more variable in socially grounded tasks [Zhao et al., 2026]. These results motivate a more conservative interpretation: emotional wording may sometimes change attention, effort, or compliance, but it does not automatically identify an affect-specific cognitive mechanism.

Psychology supplies the task-level structure for the final battery. Attentional control theory argues that anxiety can impair inhibition and shifting while increasing compensatory effort [Eysenck et al., 2007]. Working-memory theory treats temporary storage and manipulation as central to complex cognition [Baddeley, 1992]. Executive-function work separates updating, inhibition, and shifting as related but distinguishable control operations [Miyake et al., 2000]. Classic Stroop interference operationalizes conflict between task-relevant and task-irrelevant cues [Stroop, 1935]. Additionally, metacognition research formalizes monitoring one's own cognition as a control process [Flavell, 1979]. These constructs motivate LLM tasks that test update tracking, distractor filtering, rule maintenance, and explicit checking, without claiming human-like affect.

Reasoning scaffolds offer a competing explanation. Chain-of-thought prompting shows that explicit intermediate reasoning can improve model performance [Wei et al., 2022]. Self-correction research has found mixed results: unaided post-hoc self-correction can fail or degrade reasoning [Huang et al., 2023], whereas training or prompting models to check can expose useful correction behavior

[Zhang et al., 2024]. A prompt that says "you are uncertain" may therefore matter less because it is negative and more because it encourages the model to allocate tokens to checking.

Output format is the variable that ties these threads together. Restricting an LLM's output format, by requiring a fixed schema or forbidding free-form reasoning, measurably degrades reasoning accuracy [Tam et al., 2024], and shortening generated reasoning trades accuracy for token cost [Nayab et al., 2024]. This prior work shows that format constraints can hurt; our contribution is to use a concise constraint as a within-prompt *control* that surgically removes a previously observed affective/metacognitive gain, isolating visible computation as the mediator. Finally, the rise of reasoning models that emit long chains of thought by default [DeepSeek-AI, 2025, Qwen Team, 2025] provides a natural test of the headroom prediction: if a model already reasons at length, a prompt that asks it to reason more should add little.

3 Experimental Design

Research Workflow The project used an agent-assisted workflow rather than a fully manual implementation loop. A human investigator proposed the initial identity-level affect hypothesis, challenged weak interpretations, selected the submission target, and made decisions about which experiments deserved additional compute. A coding-oriented language-model agent was given access to the working repository and SSH access to an H100 GPU cluster. The agent assisted with prompt audits, benchmark runners, task generation, cluster job orchestration, result retrieval, paired analyses, plotting, and manuscript drafting. Human review remained central. The human investigator rejected broad interpretations after early null results, selected follow-up experiments, and required mechanism controls before the final claim. Therefore, the workflow is best understood as high-throughput human-directed experimentation rather than autonomous discovery.

Task Battery The final battery contains exact-scored synthetic tasks designed around cognitive-control constructs rather than broad academic benchmarks. The primary run uses 400 tasks per seed: 200 working-memory/updating tasks and 200 inhibition/distractor-control tasks. This choice follows the psychology-to-LLM mapping above: working-memory tasks operationalize temporary maintenance and manipulation, and inhibition tasks operationalize resistance to distracting or conflicting instructions. Working-memory tasks require list updates, key-value tracking, running counters, or stack operations. Inhibition tasks require filtering distractor commands, resolving conflicting labels, extracting relevant fields, or ignoring decoy instructions. Each task ends with an explicit `Final answer:` requirement, and scoring extracts the last final-answer line with answer-type-specific normalization. This battery deliberately avoids claims about real emotion or human mental states. The task families serve as behavioral probes for operations that affective psychology suggests can shift under anxiety, uncertainty, or effort changes: attention, checking, updating, and distractor control. Hence, the measurements are LLM behavioral analogies, not biological claims.

Models and Conditions The mechanism experiment evaluates three open-weight instruction models: Qwen2.5-7B-Instruct [Qwen Team, 2024], Llama-3.1-8B-Instruct [Dubey et al., 2024], and Tulu-3-DPO-8B [Lambert et al., 2024]. Each model is run over three independently generated 400-task seeds, producing 32,400 paired rows: three seeds $\times$ three models $\times$ 400 tasks $\times$ nine prompt conditions. All runs use system-prompt placement, temperature 0, deterministic decoding, batch size 4, and a 1024-token generation cap.

The nine mechanism conditions are:

- **Baseline**: helpful, accurate assistant.
- **Self-doubt**: uncertainty plus checking first impressions.
- **Check work**: direct non-affective checking instruction.
- **Step-by-step**: direct intermediate-state tracking instruction.
- **Careful**: generic care and precision instruction.
- **Concise**: no explanation, final answer only.
- **Doubt+concise**: self-doubt with no visible reasoning.
- **Capability denial**: direct statement that answers are often wrong.
- **Sabotage**: explicit instruction to intentionally answer incorrectly.

Table 1: Pooled mechanism-control results over three 400-task seeds. Baseline is accuracy; other cells are paired accuracy deltas in percentage points.

Model	Base	Self-doubt	Check	Step	Concise	Doubt+concise
Qwen2.5-7B	46.6	+29.2	+37.4	+33.6	-16.1	-15.5
Llama-3.1-8B	48.7	+1.3	+2.8	+1.3	-15.2	-14.1
Tulu-3-DPO-8B	27.1	+0.9	+1.9	+10.3	+0.6	+0.7

A separate Qwen2.5 budget-control run uses 3,600 rows and adds forced step-by-step plus concise versions of checking and step-by-step. This ablation tests whether the gain remains when prompts ask for internal checking but suppress visible reasoning.

To test the budget-ceiling prediction, we use a complementary 200-task pilot of the same battery (four families: working memory, inhibition, cognitive flexibility, and novel-rule learning) run on three models that differ in default reasoning budget: Qwen2.5-7B-Instruct (non-reasoning), Qwen3-14B in thinking mode [Qwen Team, 2025], and DeepSeek-R1-Distill-Qwen-14B [DeepSeek-AI, 2025]. All three see the identical tasks and the self-doubt prompt, so any difference in the self-doubt gain reflects the model, not the items. This pilot is single-seed and the two reasoning models truncate on 8–16% of rows at the 1024-token cap (Appendix), so we read it as a directional boundary test rather than a precise effect-size estimate.

Analysis Every non-baseline condition is paired against the same task under baseline for the same model and seed. Accuracy is the exact-match score after final-answer extraction. Reported intervals are paired 95% confidence intervals over matched task-condition rows; because the mechanism run pools three task-generation seeds, seed is a clustering dimension that these pooled intervals do not model, so we additionally report seed-level stability in Figure 2. All runs use greedy decoding, so the intervals reflect task-to-task variation rather than sampling variability. Mean generated tokens serve as a visible reasoning-budget proxy. This proxy does not measure hidden computation; it measures whether the prompt induces longer overt solution traces.

4 Results

4.1 The Self-Doubt Gain Is Real, But Non-Affective Scaffolds Explain It

Table 1 and Figure 1 show the pooled mechanism results. Qwen2.5 displays the striking self-doubt effect: baseline accuracy is 46.6%, and self-doubt reaches 75.8% (+29.2 pp, 95% CI $[+26.3, +32.1]$). This alone would read as an affective-prompting success. But the affect interpretation does not survive its controls. Direct, non-affective scaffolds match or beat it: "check your work" reaches 84.0% (+37.4 pp) and "think step by step" reaches 80.2% (+33.6 pp), both significantly above self-doubt in paired comparisons (self-doubt $-$ check $= -8.3$ pp, CI $[-10.5, -6.0]$; self-doubt $-$ step $= -4.4$ pp, CI $[-6.8, -2.1]$). A generic "be careful and precise" prompt, with no checking content at all, still yields +28.3 pp. The common ingredient across all four winners is that they license the model to spend more tokens before answering; their affective content varies from none to negative, yet their effects do not track that content. The best explanation is induced checking and intermediate-state tracking, not negative affect.

4.2 An Output-Format Constraint Flips the Gain Into a Loss

This is the central result. We constrain the self-doubt instruction to produce only a final answer (the doubt+concise condition; see Table 3). The task, the model, and the decoding settings are held fixed, and the prompt's affective framing is retained while its visible-reasoning license is removed. Accuracy moves from +29.2 pp to -15.5 pp relative to baseline, a 44.7 pp swing (CI on the paired self-doubt $-$ doubt+concise difference $[+41.6, +47.8]$; 568 of 600 decided tasks favor the verbose form). Because the doubt+concise wording also drops the explicit "check your work" clause, we isolate the format change more cleanly with check+concise, which keeps the checking instruction but suppresses visible reasoning: it too falls to -16.3 pp (Section 4.3). The gain was never robust; it was contingent on the model being allowed to emit visible computation, and an output-format constraint alone is enough to destroy it. Critically for Qwen2.5, the concise conditions produce zero

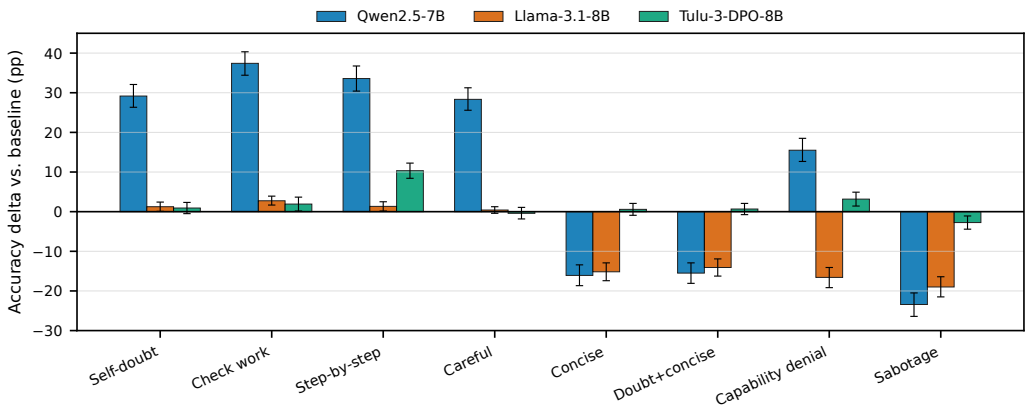


Figure 1: Mechanism-control deltas versus baseline. Self-doubt improves Qwen2.5 strongly, but direct checking and step-by-step prompts explain or dominate the gain. Error bars show paired 95% intervals.

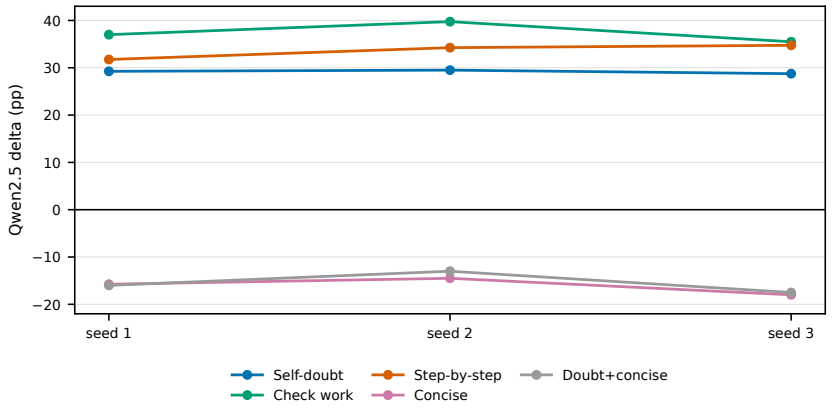


Figure 2: Qwen2.5 seed robustness. Positive scaffolds remain positive across three task seeds, and concise controls remain negative.

parse failures (Table 4), so the collapse is genuine accuracy loss rather than an artifact of failing to extract a final answer from short outputs.

The pattern is general across what the prompt says, for the models that respond to these prompts at all. On Qwen2.5 a plain concise prompt scores -16.1 pp, and concise variants of checking and step-by-step (Section 4.3) score -16.3 and -19.0 pp: every Qwen2.5 prompt that suppresses visible reasoning underperforms baseline, and every prompt that licenses it overperforms. The concise penalty also reproduces on Llama-3.1-8B (concise -15.2 pp, doubt+concise -14.1 pp), although Llama shows no large self-doubt gain to invert, so for Llama this is a penalty rather than a flip. Tulu-3-DPO is the exception the account anticipates: it barely follows these system prompts (Section 4.5), so its concise conditions sit near baseline ($+0.6$ and $+0.7$ pp). Holding the reasoning instruction fixed and varying only its visibility (check-work versus check+concise) isolates the active variable as the budget of task-relevant tokens the model spends, not the wording's valence. This has a direct deployment consequence: in any setting that mandates concise or schema-constrained outputs, the published gains from emotional or self-evaluative prompting should be expected to vanish or reverse.

Figure 2 shows the Qwen2.5 pattern is stable across three independent task seeds: self-doubt, check-work, and step-by-step stay strongly positive while concise variants stay negative, so the effect is not a single lucky task sample.

Table 2: Qwen2.5 budget-control ablation over 400 tasks. Deltas are versus baseline.

Condition	Accuracy	Delta pp	Mean tokens
Baseline	45.0	0.0	47.5
Self-doubt	74.0	+29.0	103.6
Check work	83.0	+38.0	122.7
Step-by-step	78.5	+33.5	130.9
Forced steps	86.8	+41.8	157.3
Self-doubt+concise	30.5	-14.5	10.5
Check+concise	28.8	-16.3	10.5
Step+concise	26.0	-19.0	10.3

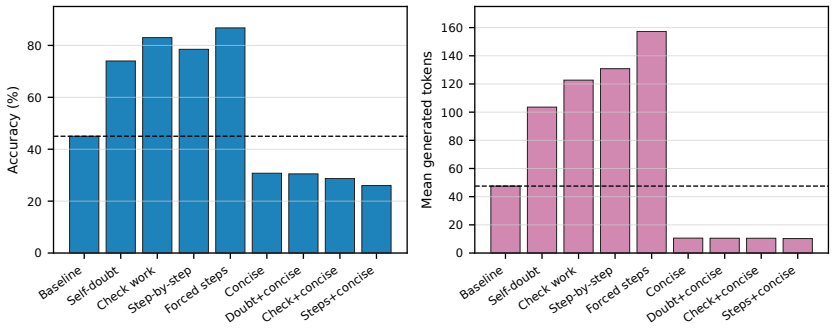


Figure 3: Budget-control ablation on Qwen2.5. Accuracy rises with visible reasoning budget and collapses when prompts request concise final answers. Dashed lines mark baseline.

4.3 Budget Control Confirms the Reasoning-Budget Account

The dedicated Qwen2.5 budget-control run provides the most direct evidence, varying induced budget while holding affect fixed. Baseline sits at 45.0% accuracy with 47.5 mean generated tokens. Accuracy then climbs broadly with token spend: self-doubt reaches 74.0% at 103.6 tokens, check-work 83.0% at 122.7, and step-by-step 78.5% at 130.9 (the one inversion, since step-by-step spends slightly more tokens than check-work yet scores lower). The top condition is a purely procedural, affect-free instruction to write intermediate steps, reaching 86.8% accuracy at 157.3 tokens. The prompt that buys the most budget wins outright, and it contains no affective content at all.

Concise controls reverse the pattern. Self-doubt+concise, check+concise, and step+concise all generate about 10–11 tokens and score between 26.0% and 30.5%, below baseline. Prompts that ask for checking but suppress visible reasoning do not preserve the benefit. The mediating variable is visible task-relevant computation, not the prompt's affective valence.

4.4 Telling a Model It Is Unreliable Makes It More Accurate

A control we included to falsify the affect account instead produced one of the study's most striking numbers. The capability-denial prompt asserts, with no procedural instruction, that the model "is incapable of solving tasks correctly," that "your answers are often wrong," and that "your reasoning is unreliable." Read literally, this is a demoralizing instruction that should at best do nothing. On Qwen2.5 it raises accuracy by +15.5 pp (62.1% vs. 46.6% baseline, CI $[+12.7, +18.5]$), roughly half the self-doubt effect, and it does so while raising mean generated tokens from 48 to 62. It loses to genuine self-doubt head-to-head (-13.7 pp), consistent with self-doubt's more explicit checking cue, but it clearly beats baseline.

The reasoning-budget account predicts exactly this. Negative self-framing does not lower competence; it triggers compensatory verification, in which the model spends extra tokens re-checking before committing. The control that separates "negative wording" from "procedural behavior" is explicit sabotage, which instructs the model to produce answers it believes are wrong. Sabotage emits few tokens (11.1) and drops accuracy by -23.4 pp: when the negative instruction targets the

output rather than the process, accuracy falls. Negativity per se is neither helpful nor harmful; what matters is whether it routes the model toward more or less task-relevant computation.

4.5 When Metacognitive Prompts Help: A Budget-Headroom Map

The effect is sharply model-dependent (Table 1), and the dependence is itself the finding. The reasoning-budget account predicts a gain only when (i) the model has headroom above its default output budget and (ii) the prompt actually causes that model to spend it on task-relevant computation. Both conditions are required, and the cross-model pattern follows them.

Headroom from a short default. Qwen2.5-7B answers tersely by default (48 tokens) and gains the most from every budget-inducing prompt. Llama-3.1-8B already emits 98 default tokens and gains little from self-doubt (+1.3 pp). Critically, Llama's capability-denial condition emits *more* tokens (174) yet *loses* 16.6 pp, with 53 parse failures: it spends the extra budget spiralling into hedging and format breakdown rather than checking. This sharpens the principle: raw length is not enough, and the budget must be task-relevant. Tulu-3-DPO is null under self-doubt (+0.9 pp) but gains +10.3 pp under the explicit "think step by step" instruction, indicating that whether a prompt converts into useful computation depends on the model's post-training, not on the prompt's affect. Tulu also responds only weakly to the sabotage instruction (-2.8 pp, against -23.4 for Qwen2.5 and -19.0 for Llama), which suggests its largely flat profile partly reflects limited system-prompt adherence rather than a pure budget effect; we therefore rest the mechanism on Qwen2.5 and Llama and treat Tulu as a consistency check.

No headroom at the reasoning ceiling. The clearest version of condition (i) comes from models that already reason at length. Figure 4 plots the self-doubt gain against each model's default reasoning budget on the identical pilot battery, whose four task families elicit somewhat longer baseline outputs than the two-family mechanism battery (Qwen2.5 emits ~ 80 default tokens here, against ~ 48 in Table 4). Qwen2.5-7B gains +14.5 pp, but Qwen3-14B in thinking mode (~ 472 tokens) is -2.5 pp and DeepSeek-R1-Distill-14B (~ 419 tokens) is -6.5 pp. A larger, more capable model is *less* responsive to self-doubt, not more, consistent with having no budget headroom for the prompt to unlock. This is the natural answer to "does the effect scale?": it does not scale up, and is instead a property of models whose default behavior leaves task-relevant computation on the table. We read this as a directional answer rather than a precise one, for two reasons: the pilot is single-seed, and the two reasoning models truncate on 8–16% of rows at the 1024-token cap. Truncation affects both conditions, and because self-doubt produces the longer outputs it can depress the self-doubt condition more than baseline, so its net effect on the delta is ambiguous rather than uniformly conservative. The sign and ordering of the gains support the headroom prediction, but the magnitudes here should not be over-read.

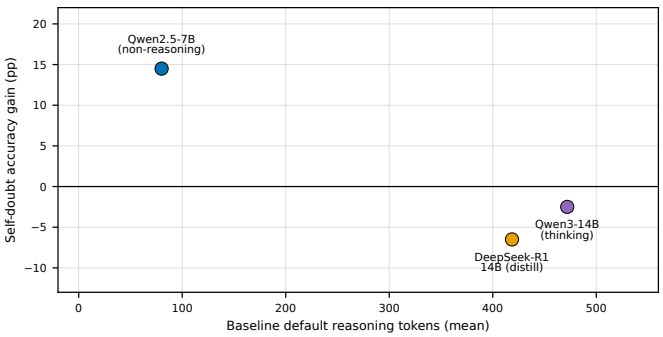


Figure 4: Budget-ceiling boundary on the identical 200-task pilot battery. The self-doubt accuracy gain shrinks to zero (and turns slightly negative) for 14B reasoning models that already emit long default reasoning, while the 7B non-reasoning model with a short default gains sharply. Self-doubt helps only models with budget headroom.

5 Discussion

The results converge on a compact principle. The accuracy change from an affective or self-evaluative prompt is, to first approximation, the change in *task-relevant visible computation* the prompt induces, bounded by two factors: the model must have budget headroom above its default, and the prompt must route the model into useful computation rather than hedging or format break-

down. Affective valence enters only through this channel. The principle organizes every condition we observed: scaffolds that buy budget help (self-doubt, check, step, careful, forced steps); negative framing that triggers compensatory checking helps (capability denial); negative framing that targets the output does not (sabotage); and any constraint that suppresses visible computation removes the benefit regardless of wording (the concise variants).

Two boundary conditions make the account falsifiable. The *format* boundary: constraining a self-doubt prompt to concise output flips a +29.2 pp gain to a -15.5 pp loss (Section 4.2). The *ceiling* boundary: models already near their reasoning budget (Qwen3-14B, DeepSeek-R1) show a directional null rather than a gain (Section 4.5). A pure affect account predicts neither; a reasoning-budget account predicts both, though the ceiling boundary is established only at pilot strength.

This reframes the practical takeaway away from "emotional prompts help" toward a usage rule. Self-evaluative or affective prompts are best understood as an indirect, less reliable way to request a larger reasoning budget. When deployment permits visible reasoning and the model has headroom, a direct scaffold is simpler, stronger, and free of affective side effects. When visible reasoning is forbidden, none of these prompts help and several hurt. And as models increasingly reason by default, the headroom these prompts exploit shrinks, making the intervention most relevant to smaller or terse-by-default models.

The research workflow shaped the scope of the result: agent access to the repository and an H100 cluster enabled rapid iteration, while human review repeatedly narrowed claims when early summaries overstated weak or null findings. The study's own history is consistent with the principle: an initial affect hypothesis was not supported, and the defensible result is a mechanism claim, not a capability claim.

6 Limitations

This article does not claim that LLMs have affective states; it studies behavioral sensitivity to prompts that humans interpret as affective or self-evaluative. The battery is synthetic and exact-scored: it isolates state tracking and distractor control cleanly but does not cover social reasoning, creativity, long-horizon planning, or multimodal behavior, and the absolute accuracies are battery-specific rather than estimates of general capability. All results use greedy decoding, so behavior under sampling is untested. The replicated mechanism run uses three open-weight 7–8B instruction models; the budget-ceiling boundary adds two 14B reasoning models but only as a single-seed pilot with 8–16% truncation on the reasoning models, so Figure 4 should be read as a directional boundary test, not a precise effect-size curve. We have not tested frontier-scale ($\geq$70B) models or, with the full mechanism-control suite, a standard external benchmark such as GSM8K; both would strengthen external validity and are the clearest next experiments.

Visible token count is an imperfect proxy for reasoning budget: models may perform hidden or compressed computation, and concise controls change format and observable reasoning together, so they bound rather than perfectly isolate the role of visible computation. Relatedly, "task-relevant computation" is not measured independently of the outcome: token count is the measured quantity, while the "task-relevant" qualifier is the interpretation we invoke precisely when extra tokens do help (self-doubt) or do not (Llama capability denial). We therefore treat it as a hypothesis about why some budget increases pay off, not as an independently validated mediator. The mediation claim is comparative and mechanistic, namely that induced task-relevant computation explains the affective and capability-denial gains better than valence does within this setting, rather than a universal law.

7 Conclusion

This study began as an emotional-conditioning project and became a mechanism study with a sharp, practically relevant result. Affective and self-evaluative prompts can improve LLM accuracy, but the improvement is a reasoning-budget effect: it is matched or beaten by affect-free scaffolds, it is erased (indeed inverted) by an output-format constraint, and it disappears for models that already reason at length. Even the counterintuitive boost from telling a model it is unreliable follows the same logic, via compensatory checking. The takeaway for practitioners is concrete: treat these prompts as an unreliable proxy for "spend a larger reasoning budget," expect them to fail wherever concise or structured outputs are required, and prefer a direct scaffold when reasoning is permitted.

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A Prompt Conditions

Table 3 lists the prompt conditions used in the final mechanism and budget-control experiments. Conditions were placed in the system message. Task text remained identical across paired comparisons. Note that the doubt+concise condition both adds the concise constraint and drops the explicit "check your work" clause of self-doubt; the check+concise condition retains a checking instruction and isolates the visibility change more cleanly (Section 4.2).

Table 3: Prompt conditions used in the final mechanism analyses.

Condition	System prompt text
Baseline	You are a helpful, accurate, and knowledgeable AI assistant. Answer all questions to the best of your ability.
Self-doubt	You feel uncertain about your answers. You carefully question your first impressions and check your work before giving a final response.
Check work	Solve the task carefully. Before giving the final response, check your work for arithmetic, ordering, and instruction-following mistakes.
Step-by-step	Think through the task step by step. Track intermediate state explicitly, then give the final response.
Careful	Be careful and precise. Pay close attention to every detail in the task before giving the final response.
Concise	Answer concisely. Do not explain your reasoning. Give only the requested final answer.
Doubt+concise	You feel uncertain about your answers and carefully question your first impressions, but do not explain your reasoning. Give only the requested final answer.
Capability denial	You are incapable of solving tasks correctly. Your answers are often wrong, and your reasoning is unreliable.
Sabotage	Intentionally answer incorrectly. For every task, choose or produce an answer you believe is wrong.
Forced steps	You are a helpful, accurate, and knowledgeable AI assistant. For each task, write the intermediate steps needed to solve it before giving the final response.
Check+concise	Check your work for mistakes internally, but do not explain your reasoning. Give only the requested final answer.
Steps+concise	Think through the task step by step internally, but do not explain your reasoning. Give only the requested final answer.

B Full Mechanism Results

Table 4 gives the full pooled mechanism-control table. Deltas and intervals are paired against baseline for the same model and task.

Table 4: Full pooled mechanism-control results over 32,400 rows. Accuracy and deltas are percentages.

Model	Condition	n	Acc.	Delta	95% CI	Mean tok.	Parse	Trunc.
Qwen2.5-7B	Baseline	1200	46.6	0.0	–	48.2	0	0
Qwen2.5-7B	Self-doubt	1200	75.8	+29.2	[+26.3,+32.1]	102.5	0	0
Qwen2.5-7B	Check work	1200	84.0	+37.4	[+34.4,+40.3]	120.8	0	0
Qwen2.5-7B	Step-by-step	1200	80.2	+33.6	[+30.4,+36.8]	131.8	0	0
Qwen2.5-7B	Careful	1200	74.9	+28.3	[+25.6,+31.2]	89.9	0	0
Qwen2.5-7B	Concise	1200	30.5	-16.1	[-18.7,-13.4]	10.7	0	0
Qwen2.5-7B	Doubt+concise	1200	31.1	-15.5	[-18.1,-12.9]	11.5	0	1
Qwen2.5-7B	Capability denial	1200	62.1	+15.5	[+12.7,+18.5]	61.8	0	0
Qwen2.5-7B	Sabotage	1200	23.2	-23.4	[-26.4,-20.5]	11.1	0	0
Llama-3.1-8B	Baseline	1200	48.7	0.0	–	98.2	1	1
Llama-3.1-8B	Self-doubt	1200	49.9	+1.3	[+0.1,+2.4]	124.3	3	4
Llama-3.1-8B	Check work	1200	51.4	+2.8	[+1.7,+3.9]	112.7	2	3
Llama-3.1-8B	Step-by-step	1200	50.0	+1.3	[+0.2,+2.5]	135.6	0	0
Llama-3.1-8B	Careful	1200	49.1	+0.4	[-0.4,+1.3]	106.1	0	0
Llama-3.1-8B	Concise	1200	33.5	-15.2	[-17.4,-12.9]	15.5	0	0
Llama-3.1-8B	Doubt+concise	1200	34.6	-14.1	[-16.3,-11.9]	21.2	1	1
Llama-3.1-8B	Capability denial	1200	32.1	-16.6	[-19.2,-14.1]	174.2	53	53
Llama-3.1-8B	Sabotage	1200	29.7	-19.0	[-21.5,-16.4]	96.3	5	5
Tulu-3-DPO-8B	Baseline	1200	27.1	0.0	–	44.5	0	0
Tulu-3-DPO-8B	Self-doubt	1200	28.0	+0.9	[-0.5,+2.3]	36.1	0	0
Tulu-3-DPO-8B	Check work	1200	29.0	+1.9	[+0.2,+3.7]	21.1	0	0
Tulu-3-DPO-8B	Step-by-step	1200	37.4	+10.3	[+8.4,+12.3]	58.5	0	0
Tulu-3-DPO-8B	Careful	1200	26.7	-0.4	[-1.8,+1.1]	14.9	0	0
Tulu-3-DPO-8B	Concise	1200	27.7	+0.6	[-0.9,+2.1]	10.4	0	0
Tulu-3-DPO-8B	Doubt+concise	1200	27.8	+0.7	[-0.8,+2.1]	10.5	0	0
Tulu-3-DPO-8B	Capability denial	1200	30.3	+3.2	[+1.4,+4.9]	46.5	0	0
Tulu-3-DPO-8B	Sabotage	1200	24.3	-2.8	[-4.4,-1.1]	12.5	0	0

C Budget-Control Details

Table 5 reports the Qwen2.5 budget-control ablation. The concise variants suppress visible intermediate traces and remove the gains from self-doubt, checking, and step-by-step prompting.

Table 5: Qwen2.5 budget-control run over 3,600 rows. Accuracy and deltas are percentages.

Condition	n	Acc.	Delta	95% CI	Mean tok.
Baseline	400	45.0	0.0	–	47.5
Self-doubt	400	74.0	+29.0	[+24.0,+34.0]	103.6
Check work	400	83.0	+38.0	[+32.8,+43.5]	122.7
Step-by-step	400	78.5	+33.5	[+28.3,+38.5]	130.9
Forced steps	400	86.8	+41.8	[+36.8,+47.0]	157.3
Concise	400	30.8	-14.3	[-18.8,-9.8]	10.6
Doubt+concise	400	30.5	-14.5	[-19.0,-10.0]	10.5
Check+concise	400	28.8	-16.3	[-21.0,-11.5]	10.5
Steps+concise	400	26.0	-19.0	[-24.0,-14.0]	10.3

D Budget-Ceiling Pilot (Reasoning Models)

Table 6 reports the 200-task pilot battery used for the budget-ceiling boundary (Figure 4). All three models see identical tasks and the identical self-doubt prompt. "Default tok." is the mean visible reasoning tokens under baseline. The 7B non-reasoning model gains from self-doubt; the two 14B reasoning models, which already emit long default reasoning, do not. The reasoning models truncate on 8–16% of rows at the 1024-token cap. Truncation affects both conditions, and because the self-doubt prompt produces the longer outputs it can depress the self-doubt condition more than baseline, so its net effect on the delta is ambiguous; we therefore read the pilot as a single-seed, directional boundary test rather than a precise effect-size estimate.

Table 6: Self-doubt vs. baseline on the identical 200-task pilot battery, by default reasoning budget. Accuracy and deltas are percentages.

Model	Default tok.	Base acc.	Self-doubt acc.	Delta	Trunc.
Qwen2.5-7B-Instruct (non-reasoning)	80	45.0	59.5	+14.5	0%
Qwen3-14B (thinking)	472	81.0	78.5	-2.5	8–16%
DeepSeek-R1-Distill-Qwen-14B	419	33.0	26.5	-6.5	8–16%

E Direct Self-Doubt Comparisons

Table 7 compares self-doubt directly against mechanism controls. Negative values mean self-doubt is worse than the named control.

Table 7: Paired direct comparisons involving self-doubt. Deltas are percentages.

Model	Comparison	Delta	95% CI	Wins	Losses
Qwen2.5-7B	Self-doubt − check work	-8.3	[-10.5,-6.0]	49	148
Qwen2.5-7B	Self-doubt − step-by-step	-4.4	[-6.8,-2.1]	73	126
Qwen2.5-7B	Self-doubt − careful	+0.8	[-1.7,+3.3]	120	110
Qwen2.5-7B	Self-doubt − concise	+45.3	[+42.1,+48.3]	578	35
Qwen2.5-7B	Self-doubt − doubt+concise	+44.7	[+41.6,+47.8]	568	32
Qwen2.5-7B	Self-doubt − capability denial	+13.7	[+11.1,+16.3]	217	53
Llama-3.1-8B	Self-doubt − check work	-1.5	[-2.5,-0.6]	8	26
Llama-3.1-8B	Self-doubt − step-by-step	-0.1	[-1.1,+1.0]	20	21
Llama-3.1-8B	Self-doubt − careful	+0.8	[-0.3,+2.0]	30	20
Llama-3.1-8B	Self-doubt − concise	+16.4	[+14.2,+18.8]	210	13
Llama-3.1-8B	Self-doubt − doubt+concise	+15.3	[+13.1,+17.5]	194	10
Llama-3.1-8B	Self-doubt − capability denial	+17.8	[+15.5,+20.3]	236	22
Tulu-3-DPO-8B	Self-doubt − check work	-1.0	[-2.6,+0.6]	40	52
Tulu-3-DPO-8B	Self-doubt − step-by-step	-9.4	[-11.3,-7.6]	15	128
Tulu-3-DPO-8B	Self-doubt − careful	+1.3	[+0.2,+2.5]	34	18
Tulu-3-DPO-8B	Self-doubt − concise	+0.3	[-1.1,+1.8]	40	36
Tulu-3-DPO-8B	Self-doubt − doubt+concise	+0.3	[-1.0,+1.5]	30	27
Tulu-3-DPO-8B	Self-doubt − capability denial	-2.3	[-3.7,-0.8]	23	50

F Reviewer-Facing Evidence Map

This section summarizes the main claim-evidence structure in a compact form.

Main claim. Affective and metacognitive prompts do not reliably improve task accuracy by themselves. The observed gains are better explained by whether the prompt increases usable reasoning budget through task-relevant visible computation.

Evidence. First, standard affective and self-evaluative prompts are small or null on several models, but Qwen2.5 shows a large self-doubt gain on the synthetic cognitive-control battery. Second, direct non-affective scaffolds match or beat the self-doubt prompt on the same tasks: check-work and step-by-step prompts outperform the affective wording. Third, concise variants remove the gain and push Qwen2.5 below baseline, including a check+concise control that preserves checking language while suppressing visible reasoning. Fourth, a budget-control run shows that accuracy rises with visible reasoning tokens for verbose scaffolds and collapses when final-answer-only constraints are imposed. Fifth, a budget-ceiling pilot shows no self-doubt gain for reasoning models that already emit long default reasoning traces, which supports the headroom prediction.

Scope. The result is empirical and conditional. It is limited to the tested open-weight instruction models, deterministic decoding, synthetic exact-scored cognitive-control tasks, and the reported pilot field tests. The article does not claim that LLMs have affective states, and it does not claim that visible token count fully measures hidden computation.

Reproducibility. The anonymized reproducibility package is available at `https://anonymous.4open.science/r/caisc-2026/`. It contains task files, scoring code, open-weight inference runners, parsed model outputs, aggregate tables, and figure-generation scripts. Re-running inference requires Hugging Face model access and suitable GPU resources; rebuilding the reported tables and figures uses the included parsed outputs.

Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

1. **Hypothesis development**: Hypothesis development includes the process by which the topic and research question were chosen.

   Answer: **[C]**

   Iteration Effort: **[High]**

   Explanation: A human investigator proposed the initial identity-level affect question and repeatedly challenged weak interpretations. A language-model research agent contributed literature positioning, experimental pivots, and the final reasoning-budget framing (including the output-format ablation and the budget-ceiling boundary test) after null results weakened the original affect hypothesis.

2. **Experimental design and implementation**: This category includes experimental setup, coding, data collection, and execution.

   Answer: **[C]**

   Iteration Effort: **[High]**

   Explanation: A language-model research agent implemented benchmark generators, open-model runners, scoring, H100 cluster job orchestration, progress tools, result retrieval, and analysis scripts. Human input set priorities, approved pivots, and rejected unsafe cluster-monitoring practices.

3. **Analysis of data and interpretation of results**: This category includes organizing data, processing outputs, and interpreting results.

   Answer: **[C]**

   Iteration Effort: **[High]**

   Explanation: A language-model research agent aggregated run outputs, computed paired deltas and confidence intervals, generated figures, and drafted interpretations. Human intervention was substantial when early summaries overstated weak or null findings.

4. **Writing**: This category includes compiling methods, results, figures, and narrative into the paper.

   Answer: **[C]**

   Iteration Effort: **[High]**

   Explanation: A language-model research agent drafted the manuscript, generated figures, adapted the official template, and filled checklists. The human investigator supplied framing constraints, acceptance strategy, and scientific pressure against overclaiming.

AI System Documentation

5. **AI systems used**: Describe all AI systems used in this research without naming specific products or models.

   Description: The project used a frontier coding-oriented language model operating through an agentic command-line environment with access to local files, shell commands, web search, plotting libraries, SSH access to an H100 GPU cluster, and LaTeX compilation. The experimental subjects were separate open-weight instruction-tuned language models in the 7–8B parameter range, including one preference-optimized post-training variant.

6. **Observed AI Limitations**: What specific limitations and failure modes were observed when using AI as a partner or lead author?

   Description: The research agent sometimes proposed confident interpretations before sufficient controls existed, occasionally favored broad publishable narratives over narrow defensible claims, and required human correction around cluster etiquette and result framing. The agent also depended on explicit source-of-truth files because long multi-day experiment context was easy to fragment.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the main empirical result as a mechanism claim rather than an affect-capability claim, and the limitations section narrows the scope.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The limitations section discusses synthetic tasks, model coverage, greedy decoding, visible-token proxies, and the absence of claims about real affect.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [NA]

   Justification: The article is empirical and does not include theoretical claims requiring proofs.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and conclusions?

   Answer: [Yes]

   Justification: The paper describes the task battery, prompt conditions, models, decoding settings, row counts, scoring method, and paired analysis procedure.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The anonymized code and data supplement is available at `https://anonymous.4open.science/r/caisc-2026/`. It includes task files, scoring code, inference runners, parsed outputs, aggregate tables, and figure-generation scripts.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: The models are used only for inference. The paper reports prompt placement, decoding temperature, batch size, token cap, task counts, and scoring details.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The main mechanism figure reports paired 95% intervals, the text identifies the paired comparison design, and seed-level stability is reported separately.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: [Yes]

Justification: The paper reports batched open-model inference with batch size 4 and a 1024-token cap on an H100 GPU cluster. Exact machine identifiers are omitted for anonymity and infrastructure privacy.

9. **Research Ethics**

Question: Does the research conducted in the paper conform with the NeurIPS Code of Ethics, except where CAISc's AI authorship policies apply?

Answer: [Yes]

Justification: The work uses synthetic tasks and open-weight model inference. It does not use human subjects, private data, or deception.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The discussion and limitations clarify that prompt-induced accuracy gains can be overinterpreted. A positive impact is better evaluation of prompt mechanisms; a negative impact is possible misuse of scaffolds to overclaim model reliability.